Credit Card Fraud Detection

AI-Powered Real-Time Transaction Analysis

Domain	Dataset	Duration
Financial Services & Fintech	284,807 Transactions	22 Days | Feb–Mar 2026

The Problem: Credit Card Fraud at Scale

$32.34B
Annual Global Fraud Losses

0.17%
Fraud Rate in Dataset

577:1
Class Imbalance Ratio

<300ms
Real-Time Decision Required

Key Challenges:

⚠ Extreme class imbalance — only 1 in 578 transactions is fraud

⏱ Real-time inference required (<300ms at point of sale)

🔄 Fraud patterns constantly evolve (concept drift)

🚫 False positives harm customer experience and revenue

Why AI Over Rules?

- ✓ Learns from 200+ variables simultaneously
- ✓ Adapts to new fraud patterns automatically
- ✓ Probabilistic scoring, not binary rules
- ✓ Explainable with SHAP (regulatory compliance)
- ✓ Scales to millions of transactions/day

Stakeholder Ecosystem

Cardholders

Primary victims of fraud

Need: Security + seamless experience

Issuing Banks

Bear financial fraud liability

Need: Loss minimization

Merchants

Accept card payments

Need: Reduce chargebacks

Payment Networks

Visa / Mastercard processing

Need: Network trust & reputation

Regulators

RBI, ECB, PCI-DSS

Need: Consumer protection

Fraud Analysts

Review flagged transactions

Need: Accuracy + efficiency

Dataset & ML Approach

Dataset Details

Source	Kaggle / ULB Group
Transactions	284,807 total
Fraud Cases	492 (0.17%)
Features	V1–V28 (PCA) + Amount + Time
Period	2 days, Sept 2013
Missing Values	None ✓

ML Pipeline

1. **Feature Engineering**
   Hour_sin/cos, Log_Amount
2. **Train/Val/Test Split**
   Stratified 64/16/20
3. **StandardScaler**
   Fit on train only (no leakage)
4. **SMOTE Oversampling**
   0.17% → 10% fraud rate
5. **XGBoost Training**
   Early stopping, AUC-PR metric
6. **SHAP Explainability**
   Top features per prediction

4 Models Evaluated — XGBoost Wins

Logistic Regression

Baseline

AUC-ROC

0.971

F1-Score

0.734

Speed

⚡⚡⚡⚡⚡

Explain.

★★★★★

Random Forest

Ensemble

AUC-ROC

0.984

F1-Score

0.852

Speed

⚡⚡⚡

Explain.

★★★

BEST

XGBoost

SELECTED

AUC-ROC

0.992

F1-Score

0.899

Speed

⚡⚡⚡⚡

Explain.

★★★★

Autoencoder

Anomaly

AUC-ROC

0.960

F1-Score

0.781

Speed

⚡⚡

Explain.

★★

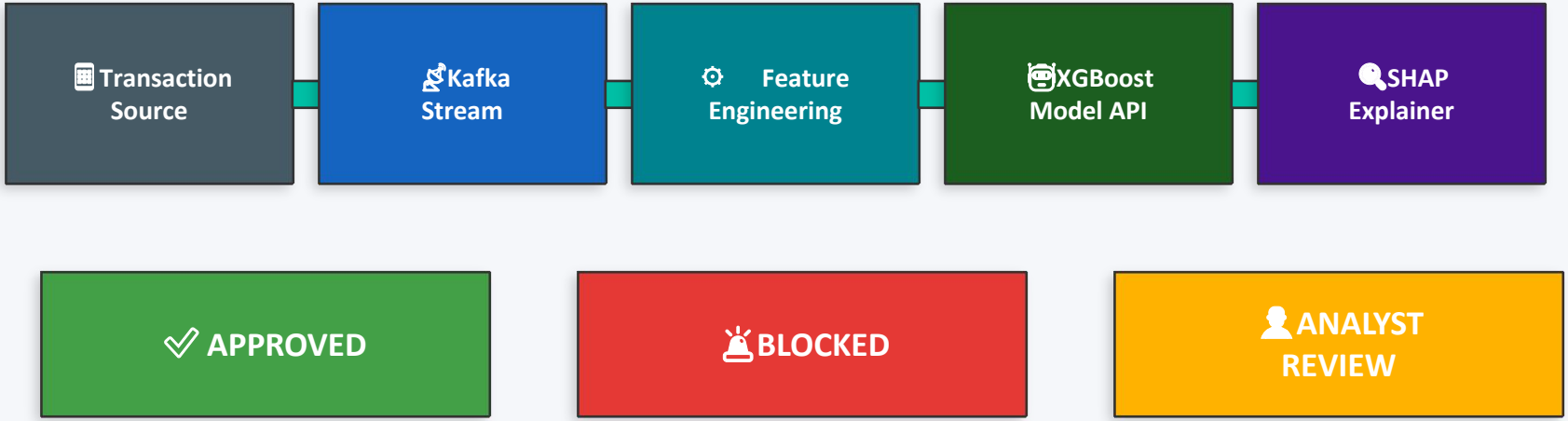
XGBoost Results & Performance

Model Performance on Test Set

Key Findings

- XGBoost outperforms all 3 competitors
- SMOTE improved recall by +22% vs baseline
- V14, V12, V4 are top SHAP features
- Optimal threshold = 0.30 (cost-sensitive)
- Inference time: ~4ms per transaction
- 85%+ fraud caught vs 62% for rule-based
- Model should retrain monthly (drift)

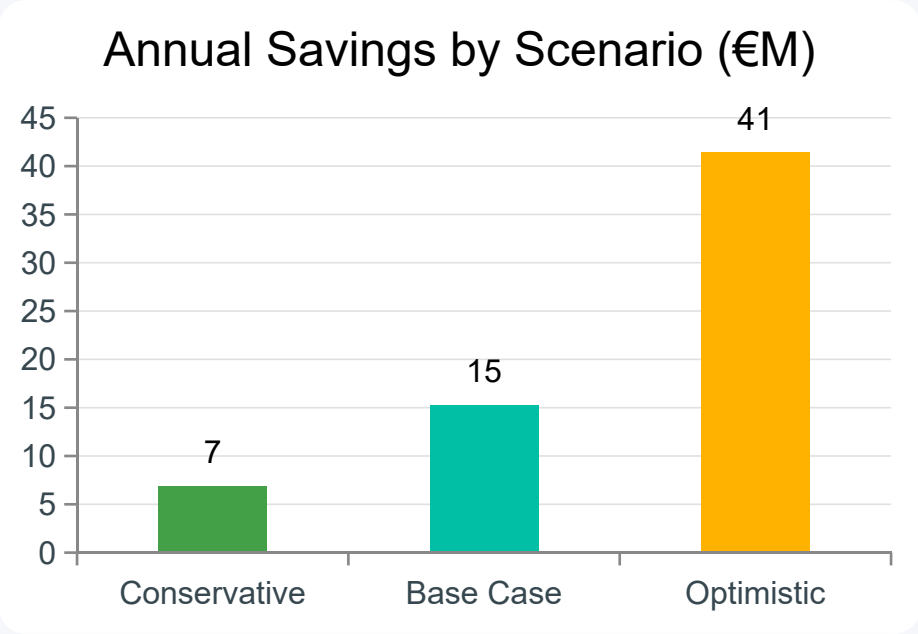
System Architecture



Technology Stack:

- Python 3.10
- XGBoost 2.0
- FastAPI
- Kafka
- Redis
- Docker/K8s
- MLflow
- SHAP
- Streamlit

Business Impact & ROI



13:1
Conservative ROI

83%
Fraud Detection Rate

€150
Avg Fraud Prevented

<50ms
Inference Latency

Implementation Cost: ~€540K/year
Break-even: < 3 months

Challenges, Ethics & Risk

Technical Challenges

Class Imbalance
- SMOTE + scale_pos_weight

Concept Drift
- Monthly retraining pipeline

False Positives
- Threshold tuning + analyst review

Real-time (<50ms)
- Redis caching + optimized model

Model Degradation
- Grafana monitoring + alerts

Ethics & Compliance

- GDPR: Data minimization & right to erasure
- PCI-DSS: Encrypted storage & transmission
- SHAP explainability for regulatory review
- Human-in-the-loop for borderline decisions
- Regular fairness audits (no demographic bias)
- Zero-trust architecture for data access

Key Takeaways & Next Steps

01

XGBoost is the Clear Winner

AUC-ROC 0.992, F1 0.899 — significantly better than rule-based systems

02

Class Imbalance is Critical

SMOTE oversampling improved recall by +22%. Never ignore data imbalance.

03

Explainability Matters

SHAP values enable regulatory compliance and analyst trust

Roadmap →

- Real-time Kafka streaming integration
- Graph Neural Networks for linked-account fraud rings
- Multi-region global deployment
- Federated learning across partner banks